

Uncertainty Propagation Using Conditional Random Fields in Large-Eddy Simulations of Scramjet Computations

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The development of scramjet engines is crucial for attaining efficient and stable propulsion under hypersonic flight conditions. Design for well-performing scramjet engines requires accurate flow simulations in conjunction with uncertainty quantification (UQ). We advance computational methods in bringing together UQ and large-eddy simulations for scramjet computations, with a focus on the HIFiRE Direct Connect Rig combustor. In particular, we perform uncertainty propagation for spatially dependent *field* quantities of interest (QoIs) by treating them as random fields, and numerically compute low-dimensional Karhunen-Loève expansions (KLEs) using a finite number of simulations on non-uniform grids. We also describe a formulation and procedure to extract *conditional* KLEs that characterize the stochasticity induced by uncertain parameters at given designs. This is achieved by first building a single KLE for each QoI via samples drawn *jointly* from the parameter and design spaces, and then leverage polynomial chaos expansions to insert input dependencies into the KLE. The ability to access conditional KLEs will be immensely useful for subsequent efforts in design optimization under uncertainty as well as model calibration with field variable measurements.

I. Nomenclature

$\sim \mathcal{N}(0, 1)$	=	Gaussian distribution with mean 0 and variance 1
A	=	matrix in regression problem
a_1	=	primary injector angle, degrees
a_c	=	scaling coefficient
a_h	=	shift coefficient
B	=	matrix in singular value problem
b	=	right-hand-side vector in regression problem
b_β, c_γ	=	polynomial chaos coefficients
C_R	=	modified Smagorinsky constant
c	=	coefficient vector in regression problem
d	=	injector diameter, m
f_{NO}	=	ratio between volume fractions of nitrogen and oxygen in the oxidizer streams
I_i, I_f	=	inlet and fuel turbulence intensity magnitudes
$\mathcal{I}, \mathcal{J}$	=	index set
$K(\cdot, \cdot), K$	=	covariance kernel and matrix
k_t	=	truncation index
L_i, L_f	=	inlet and fuel turbulence length scales, m
M	=	Mach number
M_0, M_f	=	inlet and fuel Mach numbers

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N	=	number of samples
n_d	=	number of design variables
n_g	=	number of grid points
n_p	=	number of uncertain parameters
n_t	=	number of expansion terms
Pr_t	=	turbulent Prandtl number
p	=	polynomial degree
p_0	=	inlet stagnation pressure, Pa
p_j	=	j th physical input variable
Q	=	eigenvector matrix
$q_k(\cdot), q_k$	=	k th eigenfunction and eigenvector
R_i	=	ratio of turbulence intensity vertical to horizontal components
S	=	sample matrix
Sc_t	=	turbulent Schmidt number
T	=	temperature, K
T_0	=	inlet stagnation temperature, K
U	=	left-singular matrix
V	=	right-singular matrix
$\mathcal{V}$	=	volume of computational domain
w_j, W	=	weight for the j th point, weight matrix
x, y	=	streamwise and wall-normal coordinates, m
$x/d, y/d$	=	streamwise and wall-normal coordinates normalized by injector diameter
$\mathbf{x}$	=	vector of spatial coordinates, m
x_1	=	primary injector x -location, m
x_2	=	secondary injector x -location, m
Y_{CO}	=	carbon monoxide mass fraction
β, γ	=	multi-index
δ_{kl}	=	Kronecker delta
ζ_k	=	random variables in Karhunen-Loève expansion
Λ	=	eigenvalue matrix
λ_k	=	k th eigenvalue
μ	=	regularization parameter
ξ_d	=	polynomial chaos expansion random variables corresponding to design variables
ξ_j	=	j th polynomial chaos expansion random variable
Σ	=	singular value matrix
Φ	=	standard Gaussian cumulative distribution function
ϕ_G	=	global equivalence ratio (total of primary and secondary)
ϕ_R	=	ratio of primary to secondary injector equivalence ratios
Ψ_β, Ψ_γ	=	multivariate orthonormal polynomial basis functions
ψ_{β_i}	=	univariate orthonormal polynomial basis functions
ω	=	sample space random event

II. Introduction

Research in powered hypersonic flight has thrived in the past decades with strong interests from both military and civilian aerospace applications [1, 2]. Among others, one significant technical challenge involves the need for propulsion systems that can sustain operations under hypersonic flight conditions (Mach > 5 and typically below low Earth orbit) in an efficient and stable manner [2]. In these environments, one type of air-breathing propulsion systems known as *supersonic combustion ramjet (scramjet)* engines seeks to burn fuel using atmospheric air at supersonic speeds, and thus sidesteps the necessity for carrying on-board oxidizer or decelerating airflow to subsonic speeds. Scramjets can potentially achieve much higher efficiencies compared to traditional technologies such as rockets or turbojets.

The design for well-performing scramjet engines is still in its early stages. It faces major scientific difficulties in characterizing and predicting combustion properties for multiscale and multiphysical turbulent flows under extreme conditions, and whereby mixing and combustion must occur at time scales on the order of milliseconds while flow in the

combustion chamber is supersonic. Designing an optimal engine typically involves maximizing combustion efficiency while minimizing pressure losses, thermal loading, and the risk of unstart and flame blow-out. Achieving this while also producing designs that are robust and reliable against uncertainty and noise poses an extremely challenging undertaking. An important step for advancing scramjet design therefore encompasses conducting accurate flow simulations together with uncertainty quantification (UQ).

While UQ in general has received substantial attention in the past decades, UQ for scramjet applications is much less developed but gaining traction in recent years [3–11]. A comprehensive assessment of uncertainty in such systems has been prohibitive due to the high cost of simulating turbulent reacting flows compounded with the multi-query nature of UQ investigations that generally requires some form of exploration in the stochastic space. These challenges are exacerbated when one wishes to carry out large-eddy simulations (LES), which, although more computationally demanding, can describe detailed turbulent flows and features often not accessible through the more widely used Reynolds-averaged Navier-Stokes (RANS) models. Indeed, one of several recognized computational grand challenges of LES for scramjets is “the quantification of associated uncertainties in the computational results with regard to both aleatoric and epistemic errors in input simulation parameters and physical models” [2].

We advance computational methods in bringing together UQ and LES for scramjet computations. While our previous work [6–11] explored different aspects of this endeavor, most of them studied an initial unit test problem focusing on the jet-in-crossflow physics with reaction disabled. This paper presents new results with combustion enabled. We set the scope of our UQ investigation to performing uncertainty propagation for spatially dependent *field* quantities of interest (QoIs). We treat these QoIs as random fields (RFs), and numerically compute Karhunen-Loève expansions (KLEs) (see e.g., [12]) using a finite number of simulations on non-uniform grids. KLEs are advantageous compared to directly discretizing the RFs on a grid since KLEs can effectively capture correlations across spatial dimensions while significantly reducing the stochastic degrees of freedom (DOFs). Truncated KLEs are employed in practice, where only a small number of terms are retained to capture a targeted threshold for the RF variance. We emphasize the importance of constructing KLEs using singular value decomposition (SVD), which can bring substantial memory savings when the number of simulation runs is much smaller than the number of grid points.

We also describe a formulation and procedure to extract *conditional* KLEs that characterize the stochasticity induced by uncertain parameters only, at given design conditions. A naive approach may entail repeating the KLE construction multiple times at select design points followed by interpolation, which is computationally burdensome. Instead, we build a single KLE for each QoI via samples drawn *jointly* from the parameter and design spaces, and leverage polynomial chaos expansions (PCEs) to insert input dependencies into the KLE [12, 13]. Once the joint KLE is constructed, we can then assess statistical properties and generate samples of the QoI *at different designs*. These conditional KLEs will be useful as statistically consistent surrogate models for subsequent efforts in optimal experimental design and design optimization under uncertainty where field QoIs enter the formulations. The key contributions of this paper are:

- 1) to illustrate advances in bringing together UQ and LES for scramjet computations;
- 2) to perform uncertainty propagation for spatially dependent field QoIs using RF treatments and low-dimensional KLE representations;
- 3) to demonstrate the formulation and procedure for extracting conditional KLEs that characterize the stochasticity induced by uncertain parameters only, at given design conditions.

This paper is organized as follows. Section III describes the physical problem setup and LES solver. We then introduce KLEs in Sec. IV for representing field QoIs, including their numerical construction from samples and the procedure for extracting conditional KLEs. Discussions of simulation results are presented in Sec. V. The paper concludes with a summary in Sec. VI.

III. Large-Eddy Simulations for the HIFiRE Direct Connect Rig

We concentrate on a scramjet configuration studied under the HIFiRE (Hypersonic International Flight Research and Experimentation) program [14, 15], which has been the target of a mature experimental campaign through its HIFiRE Flight 2 (HF2) project [16–18]. The HF2 payload, depicted in Fig. 1(a), consists of a cavity-based hydrocarbon-fueled dual-mode scramjet, and was tested under flight conditions of Mach 6–8+. A ground test rig, designated the HIFiRE Direct Connect Rig (HDCR) (Fig. 1(b)), was developed to duplicate the isolator/combustor layout of the flight test hardware, and to provide ground-based measurements for comparisons with flight test data, verifying engine performance and operability, and designing fuel delivery schedule [19, 20]. While data from flight tests are not publicly released, HDCR ground test data are available [19, 21]. Therefore, we aim to simulate and assess reactive flows inside the HDCR, and intend to leverage existing experimental datasets to drive modeling developments in the future.

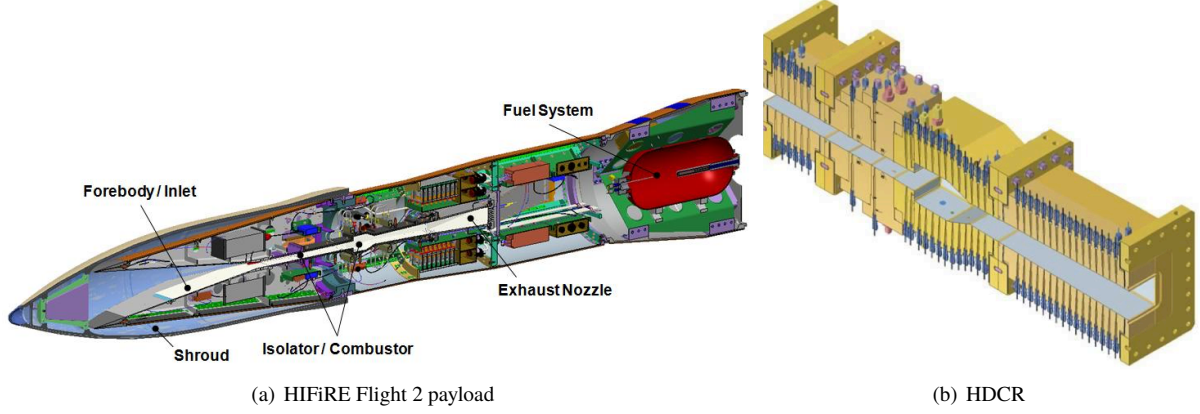


Fig. 1 HIFiRE Flight 2 payload [17] and HDCR cut view [19].

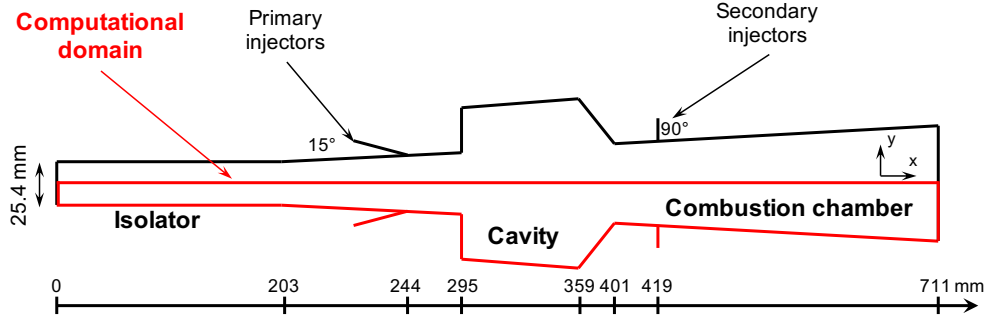
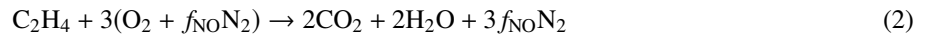
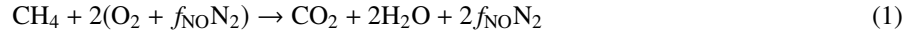


Fig. 2 Schematic of our HDCR computational domain.

The computational domain for the HDCR is highlighted by the red lines in Fig. 2. The rig consists of a constant-area isolator (planar duct) attached to a combustion chamber. It includes primary injectors mounted upstream of flame stabilization cavities on both the top and bottom walls, and secondary injectors downstream of the cavities. Flow travels from left to right in the x -direction (streamwise), and the geometry is symmetric about the centerline in the y -direction. We take advantage of this symmetry in our numerical simulations by considering a domain that covers only the bottom half of the rig. The fuel supplied through the injectors is a gaseous mixture containing 36% methane (CH_4) and 64% ethylene (C_2H_4) by volume, and acts as a surrogate with similar combustion properties as JP-7 [22]. A reduced, three-step mechanism [23, 24] is initially employed to characterize the combustion process:



where $f_{\text{NO}} = 0.79/0.21$ is the ratio between volume fractions of N_2 and O_2 in the oxidizer streams. Arrhenius formulations of the kinetic reaction rates are adopted, and the parameters are selected to retain robust/stable combustion in the current simulations.

LES calculations are then performed using the RAPTOR code framework developed by Oefelein [25, 26]. The solver has been optimized to meet the strict algorithmic requirements imposed by the LES formalism. The theoretical framework solves the fully coupled conservation equations of mass, momentum, total-energy, and species for a chemically reacting flow. It is designed to handle high Reynolds number, high-pressure, real-gas and/or liquid conditions over a wide Mach operating range while accounting for detailed thermodynamics and transport processes at the molecular level. RAPTOR is designed specifically for LES using non-dissipative, discretely conservative, staggered, finite-volume differencing. This eliminates numerical contamination of the subfilter models due to artificial dissipation and provides discrete conservation of mass, momentum, energy, and species, which is imperative for high quality LES.

Lastly, we would like to point out some limitations of our numerical results in the current paper stemming from additional simplifications necessitated by practical considerations. In particular, constraints on computational resources compelled a current UQ investigation involving simulations in a two-dimensional geometry, where we placed a single cell in the spanwise direction. (Typical computational setups and times for these runs are reported in Sec. V.) We acknowledge the decreased fidelity of these runs as a result of the reduced geometric description as well as the relatively simple chemical model in Eqn. (1) to (3). Indeed, certain physical features and phenomena are eroded or otherwise not representable in a two-dimensional setting. Nonetheless, given the scale of computations demanded by uncertainty assessments, performing UQ even under these emulatory settings has not been achieved previously. At the same time, fully three-dimensional simulations are computationally possible but only for relatively coarse grids and where only a very small number of runs can be completed under the present computational budget; they are thus not ready to support meaningful UQ analysis, and are part of our future work.

IV. Conditional Karhunen-Loève Expansions for Random Fields

A. General Formulation

We seek to construct KLEs for spatially dependent field QoIs computed from the RAPTOR LES code described in Sec. III. We begin by making a brief introduction of KLEs in the general setting without making the distinction of conditioning; mathematical details can be found in references such as [12]. A field QoI, for example temperature $T(\mathbf{x}, \omega)$, is treated as a RF that depends on both the spatial coordinates $\mathbf{x} \equiv (x, y) \in \mathcal{X}$ and sample space random event ω which induces the stochasticity of the RF (we will make this more precise when we discuss the conditioning in Sec. IV.C). Without loss of generality, we restrict ourselves to centered (zero mean) RF where, for example, the mean behavior is subtracted out from dataset samples during preprocessing. When the RF is square integrable and continuous in the mean square sense, it can be expressed via a spectral decomposition known as the KLE:

$$T(\mathbf{x}, \omega) = \sum_{k=1}^{\infty} \sqrt{\lambda_k} q_k(\mathbf{x}) \zeta_k(\omega), \quad (4)$$

where $\zeta_k(\omega)$ are mutually uncorrelated random variables with zero mean and unit variance. $\lambda_k \geq 0$ and $q_k(\mathbf{x})$ are eigenvalues and eigenfunctions, respectively, of the RF covariance function between two locations $\mathbf{x}_i$ and $\mathbf{x}_j$:

$$K(\mathbf{x}_i, \mathbf{x}_j) \equiv \mathbb{E} [T(\mathbf{x}_i, \omega), T(\mathbf{x}_j, \omega)], \quad (5)$$

and they are solutions to the homogeneous Fredholm equation of the second kind:

$$\int_{\mathcal{X}} K(\mathbf{x}_i, \mathbf{x}_j) q_k(\mathbf{x}_j) d\mathbf{x}_j = \lambda_k q_k(\mathbf{x}_i), \quad k = 1, 2, \dots \quad (6)$$

The eigenfunctions are orthonormal:

$$\int_{\mathcal{X}} q_k(\mathbf{x}) q_l(\mathbf{x}) d\mathbf{x} = \delta_{kl}, \quad k, l = 1, 2, \dots \quad (7)$$

where δ_{kl} is the Kronecker delta. The KLE can be shown to be optimal in the sense that a finite truncation under decreasing eigenvalues yields the minimum mean square error, which equals the sum of truncated eigenvalues. Thus, a k_t -term truncated KLE can provide an accurate representation of the RF using a finite number of random variables $\zeta_k(\omega)$, $k = 1, \dots, k_t$, where k_t is often much smaller than the number of grid points (i.e., DOFs from discretizing RF on a grid) especially when rapid spectrum decay takes place (which occurs under high correlation settings).

The construction of a RF thus involves characterizing λ_k , q_k , and ζ_k , which we will compute numerically from a dataset of samples of the targeted field QoI.

B. Computing Eigenvalues λ_k and Eigenvectors q_k

Let $\{\mathbf{x}_j\}_{j=1}^{n_g}$ represents the grid locations where the field variable is stored. Associated with these points are also quadrature weights $w_j > 0$ that reflect the local cell area (for example, a uniform grid may be assigned $w_j = \mathcal{V}/n_g$,

where $\mathcal{V}$ is the volume of computational domain). We begin by establishing the spatially discretized version of Eqn. (6) and Eqn. (7):

$$\int_{\mathcal{X}} K(\mathbf{x}_i, \mathbf{x}_j) q_k(\mathbf{x}_j) d\mathbf{x}_j = \lambda_k q_k(\mathbf{x}_i) \quad \longrightarrow \quad \sum_{j=1}^{n_g} w_j K_{ij} q_{k,j} = \lambda_k q_{k,i}, \quad i = 1, \dots, n_g \quad (8)$$

$$\int_{\mathcal{X}} q_k(\mathbf{x}) q_l(\mathbf{x}) d\mathbf{x} = \delta_{kl} \quad \longrightarrow \quad \sum_{j=1}^{n_g} w_j q_{k,j} q_{l,j} = \delta_{kl}, \quad (9)$$

for $k, l = 1, 2, \dots$, and where we adopted notations $K_{ij} \equiv K(\mathbf{x}_i, \mathbf{x}_j)$ and $q_{k,j} \equiv q_k(\mathbf{x}_j)$. With a total of $N < \infty$ samples, the covariance matrix can be forged from sample covariance of our dataset via the formula

$$K_{ij} = \frac{1}{N-1} \sum_{n=1}^N T(\mathbf{x}_i, \omega^{(n)}) T(\mathbf{x}_j, \omega^{(n)}), \quad i, j = 1, \dots, n_g, \quad (10)$$

where the RF is assumed to be centered in preprocessing and so the mean subtrahend is zero, and the superscript $(\cdot)^{(n)}$ represents the n th sample from the dataset. Equation (8) in matrix form is then

$$(KW)Q = Q\Lambda, \quad (11)$$

where $K \in \mathbb{R}^{(n_g \times n_g)}$, $W \in \mathbb{R}^{(n_g \times n_g)}$ is the diagonal quadrature weight matrix with $W_{jj} = w_j$, and Q and Λ are the right eigenvector matrix and the eigenvalue matrix, respectively. Two comments are warranted at this point. First, proper scaling of the eigenvectors is required in order to attain orthonormality with respect to W when non-uniform grids are involved, as required by Eqn. (9). Second, while $K \in \mathbb{R}^{(n_g \times n_g)}$, the number of nonzero eigenvalues (assuming all samples are unique) equals its rank, $\min(n_g, N-1)$, as a consequence of discretization and finite sampling. In practice, good accuracy can often be retained while truncating the expansion further with

$$T(\mathbf{x}, \omega) \approx \sum_{k=1}^{k_t} \sqrt{\lambda_k} q_k(\mathbf{x}) \zeta_k(\omega). \quad (12)$$

One practical rule of thumb is to select the smallest k_t such that λ_{k_t} has decayed to some fraction (e.g., e^{-1}) of the largest eigenvalue λ_1 . One may also choose the smallest k_t so that the retained variance (or energy) $\sum_{k=1}^{k_t} \lambda_k$ achieves some large percentage (e.g., 90%) of the total variance of the RF $\sum_{k=1}^{n_g} \lambda_k$, but this requires computing all λ_k . When the spectrum decay is rapid, k_t is often much smaller than either n_g or N .

Lastly, we note that the eigenvalue problem in Eqn. (11) can be solved through SVD. To form the related the SVD problem, we pre-multiply Eqn. (11) by $W^{\frac{1}{2}}$:

$$W^{\frac{1}{2}} KWQ = W^{\frac{1}{2}} Q\Lambda \quad \Rightarrow \quad \underbrace{\left(\frac{1}{\sqrt{N-1}} W^{\frac{1}{2}} S \right)}_B \underbrace{\left(\frac{1}{\sqrt{N-1}} S^T \left(W^{\frac{1}{2}} \right)^T \right)}_{B^T} \underbrace{\left(W^{\frac{1}{2}} Q \right)}_U = \underbrace{\left(W^{\frac{1}{2}} Q \right)}_U \Lambda \quad (13)$$

where $S \equiv [T(\mathbf{x}, \omega^{(1)}), \dots, T(\mathbf{x}, \omega^{(N)})] \in \mathbb{R}^{(n_g \times N)}$ is the sample matrix, and we used the symmetry property of W . With the SVD: $B = U\Sigma V^T$, the nonzero singular values in Σ are square root of nonzero eigenvalues in Λ , and the columns of U are related to the eigenvectors via $U = W^{\frac{1}{2}} Q$. With the truncation in Eqn. (12), iterative algorithms would be advantageous to target only the leading modes. Furthermore, since our study with scramjet computations deals with expensive simulations with large grids and few available simulation samples—that is, $n_g \gg N$ —working with the S matrix of size $(n_g \times N)$ involves much smaller memory requirements than the $(n_g \times n_g)$ system in Eqn. (11). We end this section by emphasizing that, while there is an equivalent SVD formulation for every eigenvalue problem and it is in fact what many eigenvalue solvers utilize under the hood, the key here is to *not* explicitly form the covariance matrix K , and only work with the smaller sample matrix S . For example, in our numerical results for the scramjet computations in Sec. V with the coarse, medium, and fine grid setups, forming K would respectively require 30 GB, 500 GB, and 8 TB of memory and hence infeasible on most computational platforms, in contrast to S respectively requiring only 500 MB, 430 MB, and 310 MB.

C. Conditional Karhunen-Loève Expansions by Representing ζ_k Using Polynomial Chaos Expansions

Once λ_k and q_k are obtained, we can recover from our dataset the N sample realizations of the uncorrelated random variables $\zeta_k(\omega^{(n)})$ for each k , through

$$\zeta_k(\omega) = \frac{1}{\sqrt{\lambda_k}} \int_{\mathcal{X}} T(\mathbf{x}, \omega) q_k(\mathbf{x}) d\mathbf{x} \quad \longrightarrow \quad \zeta_k^{(n)} = \frac{1}{\sqrt{\lambda_k}} \sum_{j=1}^{n_g} w_j T(\mathbf{x}_j, \omega^{(n)}) q_{k,j}, \quad n = 1, \dots, N. \quad (14)$$

One may use these samples, for example, to perform a kernel density estimate (KDE) for ζ_k , from which new samples can be generated and Eqn. (12) evaluated for subsequent uncertainty analysis. However, such an approach would not build an explicit connection to the uncertain parameters and design variables, i.e., the model inputs. Since ultimately we would like to access *conditional* KLEs that are functions of design variables and capture the stochasticity induced by uncertain parameters at given designs, we need to be more specific about the role of ω .

In the presentation so far, ω is a collective embodiment of all sources of stochasticity, and the dataset samples are perceived as different realizations of this process. To be more specific, in this work, our samples are generated by performing simulations at jointly randomized uncertain parameters as well as design variables, and so we wish to map ω into those two groups of contributions. One possible approach is to represent $\zeta_k(\omega)$ as finitely-truncated PCEs (detailed descriptions of PCEs can be found in references such as [12, 27–29]):

$$\zeta_k(\omega) = \sum_{\beta \in \mathcal{I}} b_{\beta} \Psi_{\beta}(\xi_1, \dots, \xi_{n_p}, \xi_{n_p+1}, \dots, \xi_{n_p+n_d}), \quad (15)$$

where b_{β} are the expansion coefficients, $\beta = (\beta_1, \dots, \beta_{n_p+n_d})$, $\forall \beta_j \in \mathbb{N}_0$, is a multi-index, $\mathcal{I}$ is some finite index set, n_p is the number of uncertain parameters, n_d is the number of design variables, ξ_j are a chosen set of independent random variables that correspond to each uncertain parameter and design variable, and $\Psi_{\beta}(\xi_1, \dots, \xi_{n_p}, \xi_{n_p+1}, \dots, \xi_{n_p+n_d})$ are multivariate polynomials of the product form

$$\Psi_{\beta}(\xi_1, \dots, \xi_{n_p}, \xi_{n_p+1}, \dots, \xi_{n_p+n_d}) = \prod_{i=1}^{n_p} \psi_{\beta_i}(\xi_i) \prod_{j=n_p+1}^{n_p+n_d} \psi_{\beta_j}(\xi_j), \quad (16)$$

with ψ_{β_i} denoting degree- β_i polynomials orthonormal with respect to the probability density function of ξ_i (i.e., $p(\xi_i)$):

$$\mathbb{E} [\psi_k(\xi_j) \psi_n(\xi_j)] = \int_{\Xi_j} \psi_k(\xi_j) \psi_n(\xi_j) p(\xi_j) d\xi_j = \delta_{kn}. \quad (17)$$

While different choices of ξ_j and ψ_{β_j} are available under the generalized Askey family [30], our application will employ Hermite expansions with Gaussian $\xi_j \sim \mathcal{N}(0, 1)$. While this choice might seem counterintuitive given our parameters and design variables are endowed with uniform measures, it is made due to ζ_k 's being observed to be near-Gaussian in our numerical results Sec. V and so Gauss-Hermite expansions are expected to provide better representations. Note that for the particular case of Gaussian random fields (not assumed here) ζ_k would be exactly standard Gaussian random variables. We thus represent the uniform distributions of parameters and designs by standard Gaussian distributions through the cumulative distribution function of a standard Gaussian (denoted by Φ here):

$$p_j = a_h + a_c \Phi(\xi_j), \quad (18)$$

where p_j denotes the j th physical input variable (either uncertain parameter or design variable listed in Table 1), and a_h and a_c are shift and scaling coefficients to establish the correct mean and range for that uniform distribution. We then adopt Gauss-Hermite expansions between these quantities and ζ_k which should now be captured with lower order polynomials compared to if uniform-Legendre PCEs were utilized. For simplicity, we focus on total-order expansions of degree p with $\mathcal{I} = \{\beta : \|\beta\|_1 \leq p\}$. The total number of basis terms for a total-order expansion is $\frac{(n_p+n_d+p)!}{(n_p+n_d)!p!}$.

Once the PCE in Eqn. (15) is available, we can use it to rapidly sample (either joint or conditional) realizations of ξ_j , which translate to samples of ζ_k and then T . We can also extract moment information (either joint or conditional) analytically from PCE coefficients. For example, the mean field of T conditioned on a specific design (that is, conditioned on $(p_{n_p+1}, \dots, p_{n_p+n_d})$, or equivalently conditioned on $\xi_d \equiv (\xi_{n_p+1}, \dots, \xi_{n_p+n_d})$ through Eqn. (18)) is

$$\mathbb{E} [T(\mathbf{x}, \omega) | \xi_d] \approx \mathbb{E} \left[\sum_{k=1}^{k_t} \sqrt{\lambda_k} q_k(\mathbf{x}) \zeta_k(\omega) \middle| \xi_d \right] = \sum_{k=1}^{k_t} \sqrt{\lambda_k} q_k(\mathbf{x}) \mathbb{E} [\zeta_k(\omega) | \xi_d]. \quad (19)$$

Similarly, the conditional variance field is

$$\text{Var}[T(\mathbf{x}, \omega)|\xi_d] \approx \text{Var}\left[\sum_{k=1}^{k_t} \sqrt{\lambda_k} q_k(\mathbf{x}) \zeta_k(\omega) \middle| \xi_d\right] = \sum_{k=1}^{k_t} \lambda_k [q_k(\mathbf{x})]^2 \text{Var}[\zeta_k(\omega)|\xi_d], \quad (20)$$

where the equality establishes from ζ_k being mutually uncorrelated. The standard deviation field can be obtained by taking a pointwise square root of the variance field. To calculate $\mathbb{E}[\zeta_k(\omega)|\xi_d]$ and $\text{Var}[\zeta_k(\omega)|\xi_d]$, we substitute the desired ξ_d values into Eqn. (15), collect like-terms in the new n_p -dimensional PCE $\zeta_k(\omega)|\xi_d = \sum_{\gamma \in \mathcal{J}} c_\gamma \Psi_\gamma(\xi_1, \dots, \xi_{n_p})$, and evaluate $\mathbb{E}[\zeta_k(\omega)|\xi_d] = c_0$ and $\text{Var}[\zeta_k(\omega)|\xi_d] = \sum_{0 \neq \gamma \in \mathcal{J}} c_\gamma^2$.

Since we have samples $\zeta_k^{(n)}$, $n = 1, \dots, N$ from Eqn. (14), we may estimate PCE coefficients c_β via regression. This involves solving c in the following regression linear system $Ac = b$:

$$\underbrace{\begin{bmatrix} \Psi_{\beta^1}(\xi^{(1)}) & \dots & \Psi_{\beta^{n_t}}(\xi^{(1)}) \\ \vdots & & \vdots \\ \Psi_{\beta^1}(\xi^{(N)}) & \dots & \Psi_{\beta^{n_t}}(\xi^{(N)}) \end{bmatrix}}_A \underbrace{\begin{bmatrix} c_{\beta^1} \\ \vdots \\ c_{\beta^{n_t}} \end{bmatrix}}_c = \underbrace{\begin{bmatrix} \zeta_k^{(1)} \\ \vdots \\ \zeta_k^{(N)} \end{bmatrix}}_b, \quad (21)$$

where the notation Ψ_{β^i} refers to the i th basis function for a total of n_t terms, c_{β^i} is the coefficient corresponding to that basis, and $\xi^{(n)}$ is the n th regression point (here ξ without subscript represents the vector of all ξ_j). The Karhunen-Loève theorem states that ζ_k are zero mean, unit variance, and mutually uncorrelated. We remove the constant basis term to impose zero mean, but we currently do not impose explicit constraints for unit variance and correlation. Enforcing those constraints is non-trivial, and warrants additional algorithmic development. A is thus the regression matrix where each column corresponds to a basis (except the constant basis term) and each row corresponds to a regression sample point. The number of columns n can easily become quite large in high-dimensional settings; for example, a total-order expansion of degree 3 in 16 dimensions contains $n_t = \frac{(3+16)!}{3!16!} - 1 = 969$ terms. In fact, we often encounter situations where the number of samples N may be much smaller than n_t . We thus employ compressive sensing [31, 32] to discover sparse structures in PCEs by finding a sparse solution for the (often underdetermined) system in Eqn. (21). Specifically, we use the unconstrained least absolute shrinkage and selection operator (LASSO) form:

$$\min_c \|Ac - b\|_2^2 + \mu \|c\|_1 \quad (22)$$

where μ is a regularization constant. We solve the LASSO problem numerically using the alternating direction method of multipliers (ADMM) [33, 34], and select μ based on cross-validation techniques in the manner utilized by [7].

V. Numerical Results

In our numerical studies, we designate 11 parameters to bear uncertainty (uncertain parameters), and allow a separate set of 5 parameters to vary and reflect the scramjet design configuration (design variables). The total model input space is thus 16 dimensional, and tabulated in Table 1. The uncertain parameters reflect uncertainty in inlet and fuel inflow boundary conditions as well as turbulence model parameters for the LES sub-grid scale model. With lower and upper bounds suggested by subject matter experts, we invoke the maximum entropy principle [35, 36] and endow these parameters with uninformative uniform “prior” distributions across the ranges indicated in the table. While there are no intrinsic distributions associated with the design variables, we view them as random variables in the exploratory stage. Without any prior information to favor particular regions of the design space, we also sample the design variables in accordance with uniform measures*.

We apply our method on multiple field QoIs, such as temperature (T), Mach number (M), mass fractions of chemical species (e.g., Y_{CO}), etc. All QoIs are time-averaged unless indicated otherwise. The simulation data utilized in the current analysis are from two-dimensional scramjet computations, employing grid resolutions where cell sizes are 1/8 (coarse), 1/16 (medium), and 1/32 (fine) of the injector diameter $d = 3.175 \times 10^{-3}$ m, corresponding to respectively around 63 thousand, 258 thousand, and 1 million grid points. The number of time steps for each run at different grid resolutions

*A more effective approach is to sample design variables proportional to the frequency or probability of where the optimizer visits, since we would like more samples (and thus higher weight on accuracy) in areas that are evaluated more often. However, it is heavily dependent on the problem and the numerical methods employed, and very difficult to assess *a priori*.

Parameter	Range	Description
Inlet boundary conditions:		
p_0	$[1.406, 1.554] \times 10^6$ Pa	Stagnation pressure
T_0	$[1472.5, 1627.5]$ K	Stagnation temperature
M_0	$[2.259, 2.761]$	Mach number
I_i	$[0, 0.05]$	Turbulence intensity horizontal component
R_i	$[0.8, 1.2]$	Ratio of turbulence intensity vertical to horizontal components
L_i	$[0, 8] \times 10^{-3}$ m	Turbulence length scale
Fuel inflow boundary conditions:		
I_f	$[0, 0.05]$	Turbulence intensity magnitude
L_f	$[0, 1] \times 10^{-3}$ m	Turbulence length scale
Turbulence model parameters:		
C_R	$[0.01, 0.06]$	Modified Smagorinsky constant
Pr_t	$[0.5, 1.7]$	Turbulent Prandtl number
Sc_t	$[0.5, 1.7]$	Turbulent Schmidt number
Design variables:		
ϕ_G	$[0.5, 0.8]$	Global equivalence ratio (total of primary and secondary)
ϕ_R	$[0.25, 0.35]$	Ratio of primary to secondary injector equivalence ratios
x_1	$[0.231, 0.2564]$ m	Primary injector x -location
x_2	$[0.40755, 0.43295]$ m	Secondary injector x -location
a_1	$[5, 25]^\circ$	Primary injector angle

Table 1 Uncertain model parameters and design variables. The distributions are assumed uniform across the ranges shown.

are selected to maintain approximately equal simulated physical times, together considering practical constraints due to computational resources. While the timestep sizes are determined adaptively based on the Courant-Friedrichs-Lewy (CFL) criterion, their values are approximately proportional to the grid spacing. The simulations are thus conducted for 2×10^5 iterations for the coarse grid, 4×10^5 for medium, and 8×10^5 for fine, from their respective warm-start solutions that were engineered from a quasi-steady state nominal condition simulation. The instantaneous solutions corresponding to the second half of each run are time-averaged to generate a final solution. The total numbers of simulations performed in establishing our database, as well as their average CPU times, are shown in Table 2.

Grid resolution	Run length (iterations)	Approximate CPU hours per run	Total number of runs
Coarse	2×10^5	1.7×10^3	1053
Medium	4×10^5	1.1×10^4	222
Fine	8×10^5	7.3×10^4	39

Table 2 Run lengths, CPU hours per run, and total numbers of runs performed in establishing our database.

Figure 3 presents the sample mean and standard deviation fields for T , M , and Y_{CO} , for the medium ($N = 222$ samples) grid (results for the coarse and fine grids are omitted for brevity). Figure 4 contains the Y_{CO} mean and standard deviation fields compared across the coarse ($N = 1053$ samples), medium ($N = 222$ samples), and fine ($N = 39$

samples) grids. The mean and standard deviation are computed from our database of runs in Table 2, which consists of simulations conducted at input samples drawn *jointly* from the parameter and design spaces in accordance to Table 1. Together with effects of time-averaging, these plots thus are not meant to display detailed time-dependent LES flow features, but rather a summary of statistical average behaviors. We emphasize that working directly with these samples cannot produce *conditional* statistics for any given design variables, which would be valuable for subsequent design optimization. We thus produce conditional RFs using the tools introduced in this paper.

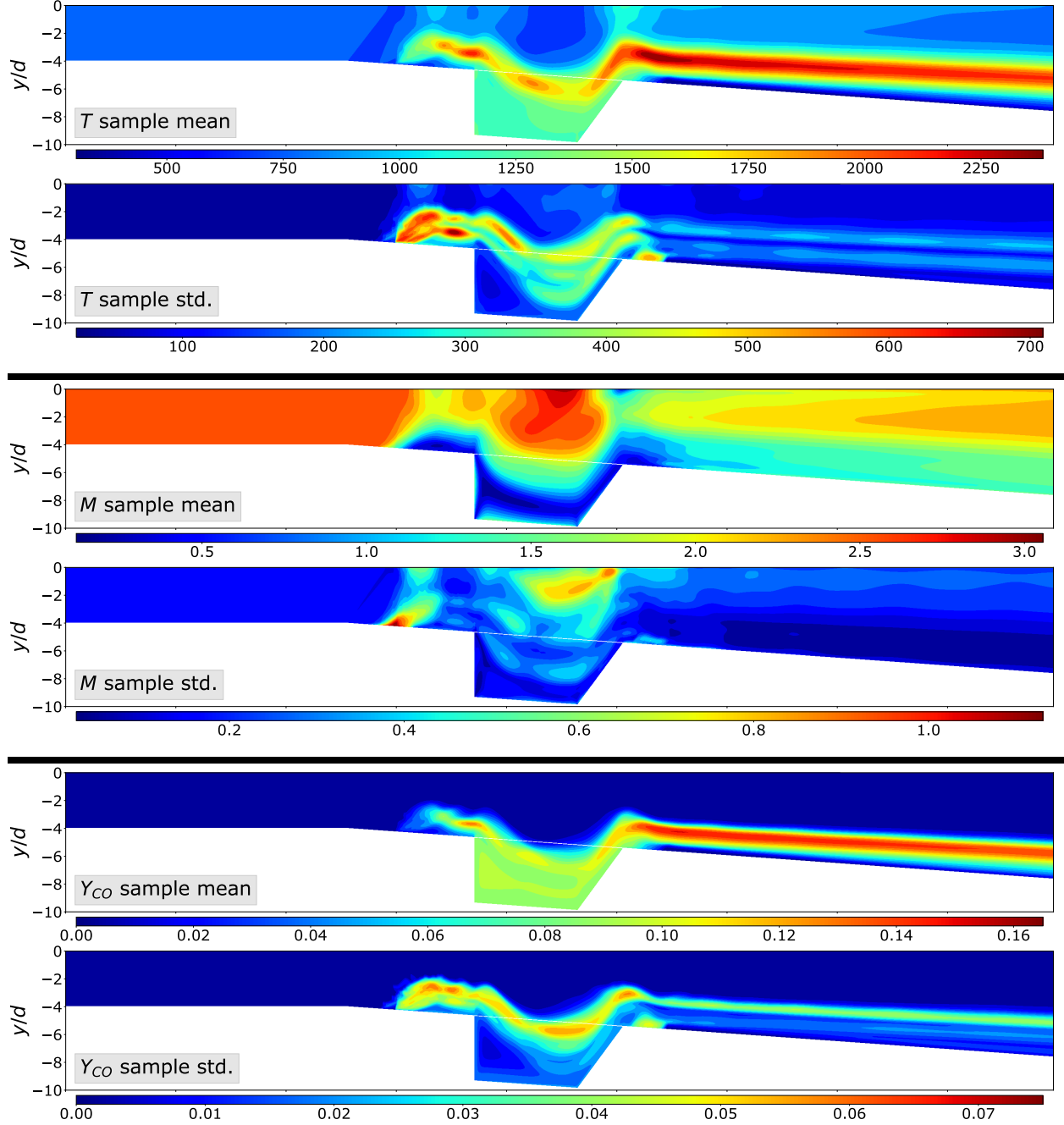


Fig. 3 Sample mean and standard deviation fields for T [K] (top two), M (middle two), and Y_{CO} (bottom two) from $N = 222$ medium grid database samples.

Following the procedure described in Sec. IV, we construct KLE for each of the three field variables considered here. Figure 5 plots the largest 100 eigenvalues λ_k for each grid resolution (dense grid has only 39 nonzero eigenvalues).

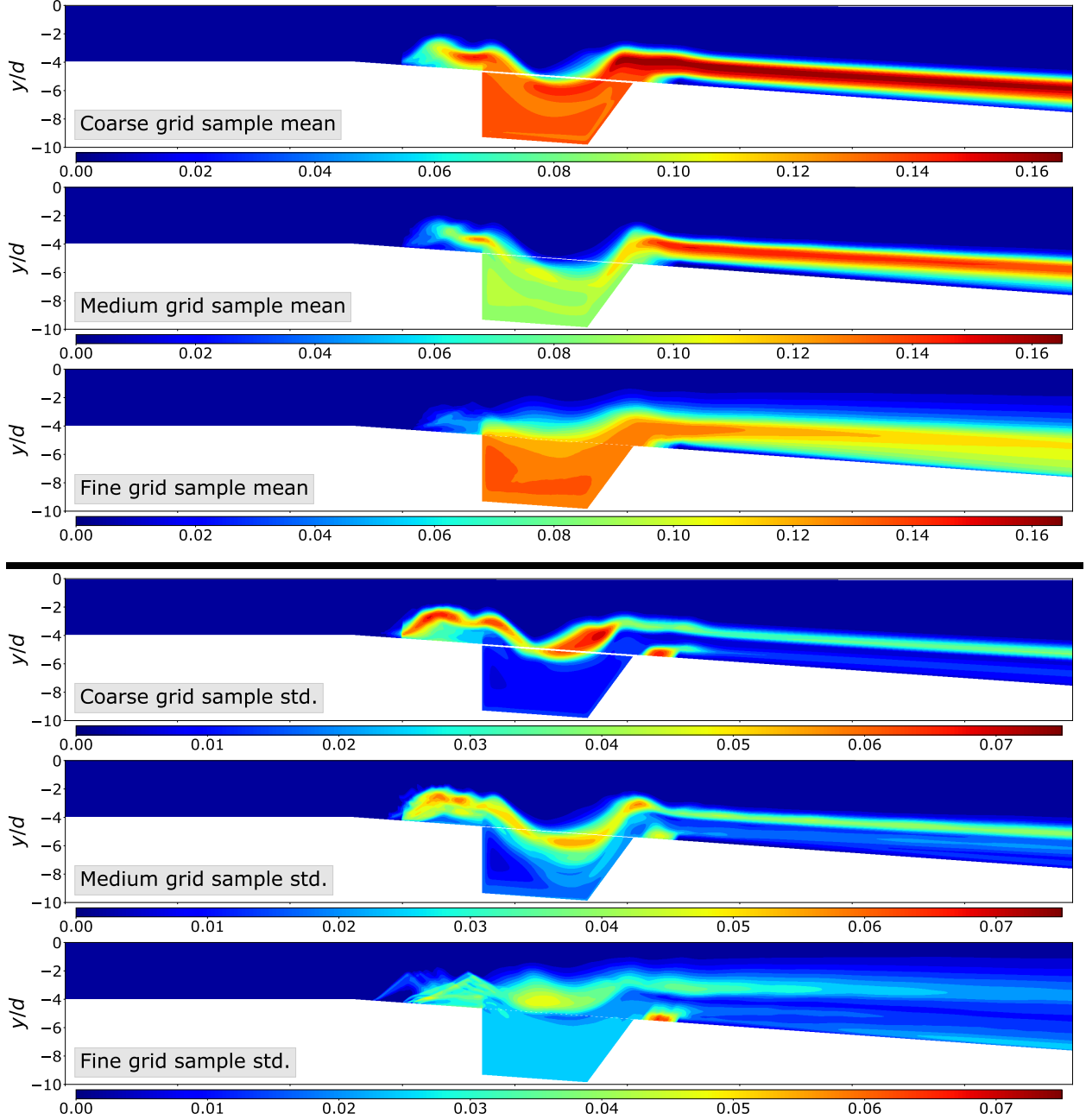


Fig. 4 Sample mean (top three) and standard deviation (bottom three) fields for Y_{CO} from $N = 1053$ coarse, $N = 222$ medium, and $N = 39$ fine grid database samples.

Rapid decay is observed in all grids and all variables, and a truncation of $k_t = 10$ arrives at eigenvalues less than 10% of the largest λ_k ; we employ $k_t = 10$ for subsequent analysis. Figure 6 illustrates the first five KLE modes for Y_{CO} , while the higher modes and other variables are omitted for brevity. To better highlight the modal response, we scale the positive and negative regions separately by their largest values in each plot, and color the positive with yellow-red contours and negative with green-blue. One may interpret the earlier modes to be the most relevant features for capturing the “big picture” of the corresponding field variable, and we indeed observe the earlier modes tend to depict the “tail” component leaving the cavity section. Overall, however, the complicated structures in this flow make it difficult to qualitatively verify the modes intuitively. We also see rather large differences between figures for the three different grid resolutions (and similarly observed in Fig. 4). While all these plots are based on statistical behavior (across the

uncertain and design variables) of the field quantities and thus are affected by the number of available runs (samples) in the database, another (probably more dominating factor) is that the grid resolutions currently employed likely have not reached the grid converging regime. More refined meshes would be a target of future investigation when more computational resources become available. Completing the KLE, Fig. 7 presents histograms on the first three ζ_k for Y_{CO} . Higher modes, and variables T and M yield similar figures and thus omitted. As expected from the Karhunen-Loève theorem, all ζ_k samples are verified to have (numerically) zero-mean, unit-variance, and are mutually uncorrelated. The histograms appear close to standard Gaussian, but some deviations are present. The near-Gaussian observation motivate the employment of Gauss-Hermite PCEs to characterize ζ_k , as discussed in Sec. IV.C. A second-order PCE is created for each ζ_k for coarse and medium grids, and first-order for fine grid since only 39 samples are available.[†]

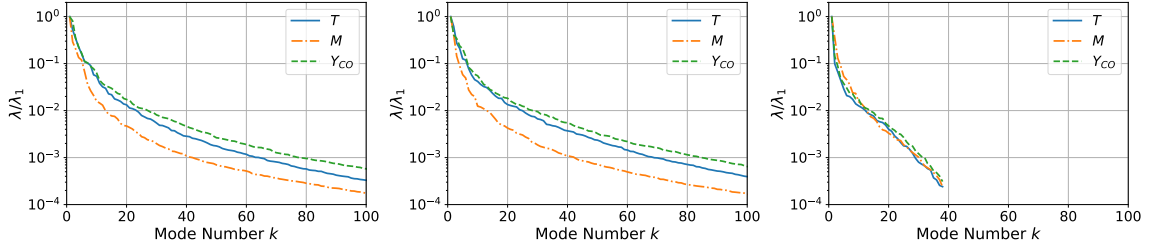


Fig. 5 Largest KLE eigenvalues for coarse (left), medium (middle), and fine (right) grids.

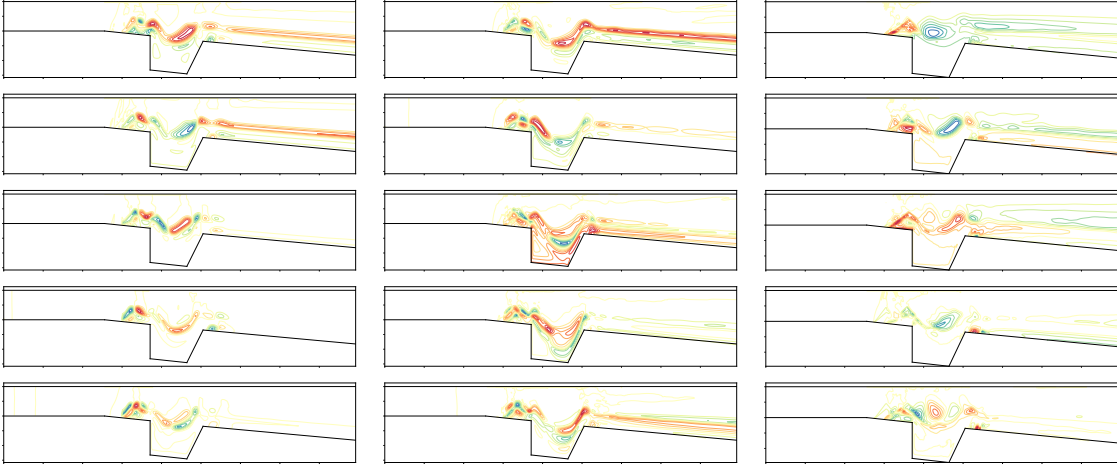


Fig. 6 Y_{CO} KLE modes for coarse (left), medium (middle), and fine (right) grids, from $k = 1$ (top) to $k = 5$ (bottom).

Lastly, with the overall KLE acting as a generative model with differentiating dependencies on different parameters and designs, we can now generate samples of the field variables rapidly, and also compute their moment statistics analytically. For demonstration, we compute the mean and standard deviation RFs conditioned at three different designs specified in Table 3: design 1 sets all design variables at their mid-values in accordance to Table 1, design 2 at the lower 25% values, and design 3 at the upper 25% values. The conditional mean and standard deviation fields for Y_{CO} using the medium grid are illustrated in Fig. 8. The conditional mean RFs illustrate the mean behavior as design configuration is altered, and we notice the differences in injector locations and fuel jet angles among the three designs. While the injector locations are consistent with the relative values between designs, the jet angles appear to conflict likely due to approximation errors (discussed below). The conditional standard deviation RFs reflect variation due to uncertain parameters only, and they exhibit values smaller than the sample standard deviation RFs in Fig. 4 that encompass variation due to both uncertain parameters and design variables. This observation is consistent with our intuition. The

[†]Note that a first-order expansion would not be able to reflect conditional standard deviation fields, since all design variable terms would contribute to the constant term (mean field) upon conditioning on a design choice, and the variance, which is computed from all other terms, would always remain the same. For the fine grid case, more runs are then need to construct at least second-order PCEs in order to access conditional RFs.

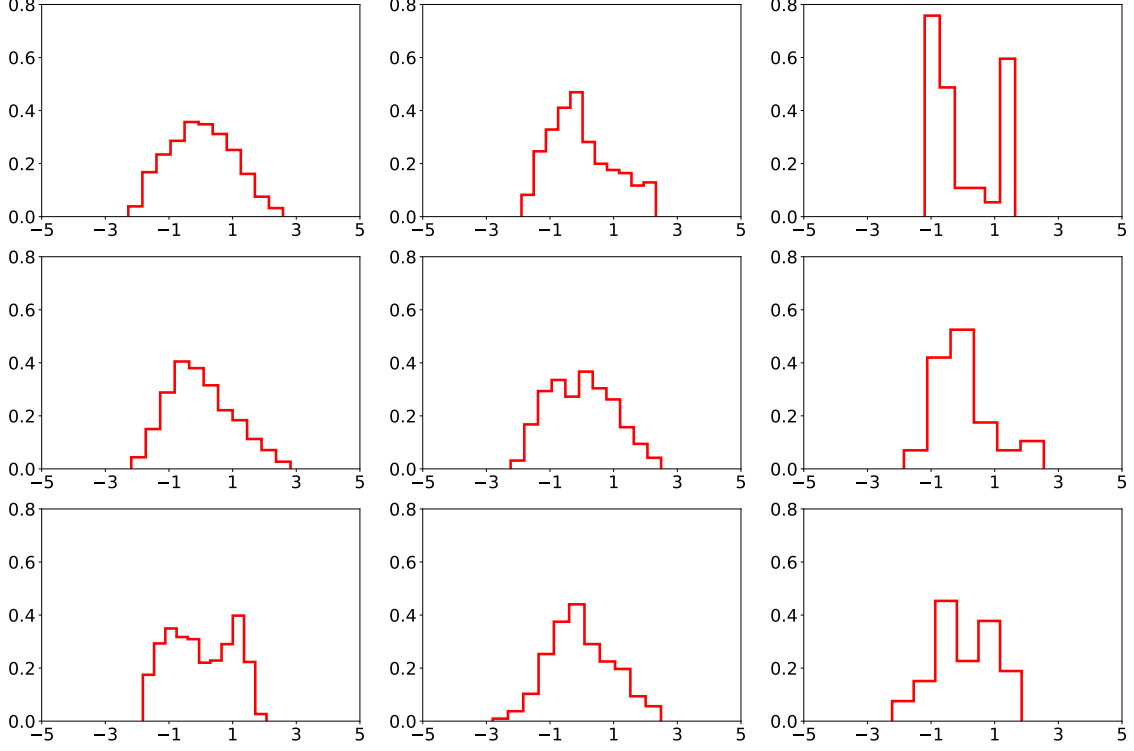


Fig. 7 Normalized histograms for ζ_k (where each sample corresponds to a run from the database) for Y_{CO} with coarse (left), medium (middle), and fine (right) grids, from $k = 1$ (top) to $k = 3$ (bottom).

conditional RFs are able to depict differences in the mean and standard deviation as the design condition is changed, but their accuracy compared to direct Monte Carlo RAPTOR simulations needs to be assessed. There are several potential sources of error: finite (and small) number of database samples, assumptions underlying the Karhunen-Loève theorem, truncation in KLE, and PCE representations of ζ_k . Future efforts for quantifying and controlling these errors with adaptive algorithms would be beneficial. We also note that constructing KLEs on transformations of the QoIs, e.g., log of QoIs, lead to high errors due to the limited number of samples available for this study.

Design No.	Name	$[\phi_G, \phi_R, x_1, x_2, a_1]$
1	“Center”	[0.65, 0.30, 0.2437, 0.42025, 15]
2	“Lower 25%”	[0.575, 0.275, 0.23735, 0.4139, 10]
3	“Upper 25%”	[0.725, 0.325, 0.25005, 0.4266, 20]

Table 3 Three design conditions selected for demonstrating conditional RFs.

VI. Conclusions

This paper presents algorithms aimed at advancing uncertainty quantification (UQ) efforts for large-eddy simulations (LES), under an application of supersonic reactive flow inside the HIFiRE Direct Connect Rig (HDCR) combustor that is important for scramjet engine research. We target spatially dependent field quantities of interest (QoIs) in LES, and treat them as random fields (RFs) with randomness induced by uncertain model parameters and design variables. We describe methods for representing these RFs via low-dimensional Karhunen-Loève expansions (KLEs), constructed numerically using simulation results from a set of randomized inputs. Once constructed, the KLEs are useful for both forward propagation of uncertainty and model calibration. However, the most useful forms require *conditional* KLEs that characterize the stochasticity induced by uncertain parameters only, at given design values. We thus further

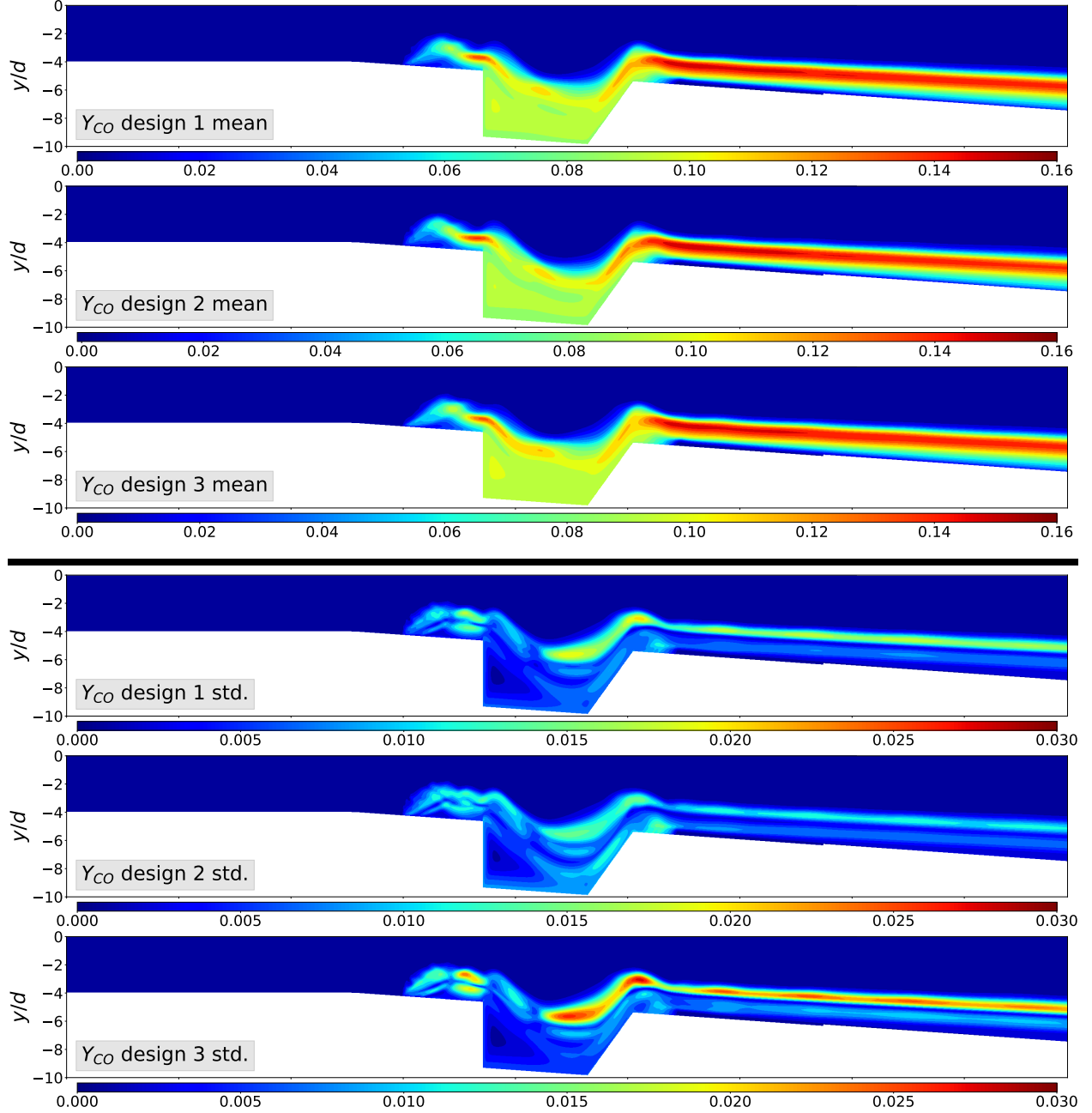


Fig. 8 Conditional mean (top three) and standard deviation (bottom three) fields at designs 1, 2, and 3 for Y_{CO} from the medium grid.

present a computational procedure to obtain these conditional RFs, by first building a single KLE for each QoI via samples drawn *jointly* from parameter and design spaces, and then leverage polynomial chaos expansions to insert input dependencies into that KLE. The resulting conditional KLEs, which are generative models at different designs, provide a computationally inexpensive alternative to support subsequent design optimization under uncertainty where field QoIs enter the formulation, as well model calibration from experimental or high-fidelity field QoI measurements.

In our numerical investigations, we find that truncated KLE models with approximately 10 modes are sufficient to capture around 90% of the overall variance for QoIs studied in this paper: temperature, Mach number, and CO mass fraction fields. We have successfully accessed conditional KLEs, along with their mean and standard deviation fields, when only samples from the joint space are available. Approximation errors need to be further assessed and controlled,

especially for expensive finer grid cases when very few simulations are available. At the same time, even the finest mesh resolution used in this study is suspected to have not reached the grid convergent regime, and it further highlights the substantial challenges of exercising UQ with LES.

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